Assignment 2: Analysis of Competing Hypotheses

Weight	10% of final grade
Format	Single PDF
Report due	November 6th, 2026 at 11:59 PM via CourseLink

1. Purpose

Lecture 5 introduced the Analysis of Competing Hypotheses (ACH) methodology (Heuer, 1999) as a structured technique for resisting confirmation bias and satisficing in intelligence analysis. This assignment asks you to apply the full methodology, independently and in depth, to a single realistic cyber intelligence problem: Aurora Biotech (assignment2_scenario.pdf).

An ACH will not generate a single correct answer but rather it promises a *defensible, transparent* answer where another analyst can see exactly which evidence drove your conclusion and can disagree with your judgment while agreeing you followed the method honestly. That is what this assignment grades: the quality and honesty of your reasoning process, not whether you land on the specific hypothesis the instructor had in mind.

2. Deliverables

Two deliverables, submitted as a single PDF.

2.1 An ACH Matrix

Following the method from Lecture 5:

1. **Hypotheses:** Start from the four hypotheses given in the scenario. You must add **at least one additional hypothesis** of your own that is not simply a reworded version of one already given — a genuine distinct explanation the evidence could support.
2. **Evidence and assumptions:** Use **all** of the evidence items provided in the scenario. You may add additional evidence or reasonable assumptions you think the analysis needs (clearly labelled as your own addition, not scenario fact), but **you may not delete or ignore any evidence item given**.
3. **Ratings:** For every evidence item against every hypothesis, rate it Consistent (C), Inconsistent (I), or Not Applicable/Diagnostic-neutral (NA), exactly as covered in lecture. Briefly justify any rating that is not obvious from a first read.
4. **Inconsistency tally:** Total the inconsistency count per hypothesis.
5. **Diagnosticity:** Identify which 2–3 evidence items are doing the most analytical work. These are the ones that discriminate most sharply between hypotheses and explain what that is the case. An item that is “Consistent” with all four hypotheses is confirming very little; say that too, if it’s true of an item on your matrix.

A template matrix structure is provided in assignment2_matrix_template.csv; you are not required to use it, but your matrix must contain the same information.

2.2 A written analytic report (1200-1800 words)

Structured with these sections, each clearly headed:

- **Key judgment** (2–4 sentences): your best-supported hypothesis, stated up front, with an explicit confidence level (e.g., “moderate confidence”) that is not buried at the end.
- **Hypothesis set**: the four given hypotheses, your added hypothesis, and one sentence on why your addition is genuinely distinct rather than a restatement.
- **Evidence and diagnosticity**: which evidence items most discriminate between hypotheses, and why. This is the heart of the assignment. Do not just summarize the matrix in prose; explain what the matrix *means*.
- **Inconsistencies and what they rule out (or don’t)**: which hypotheses accumulate the most inconsistencies, and whether a high inconsistency count genuinely disqualifies a hypothesis here or whether some of those inconsistencies are themselves explainable.
- **Conclusion, confidence, and what would change your mind**: your final judgment, stated honestly (do not overclaim certainty the evidence doesn’t support, and do not hedge so much that you haven’t actually concluded anything), plus a brief sensitivity note: what single new piece of evidence would most change your answer if it turned out to be true or false — and one or two concrete indicators you would monitor for next.

Word count is a hard requirement, not a suggestion: state your word count (excluding the matrix and headers) at the top of the report. Reports under 1,200 or over 1,800 words will be marked down under “Structure, Clarity, and Analytic Writing” on the rubric regardless of content quality.

4. Submission Format

- **Format:** Upload a single PDF to the Assignment 2 dropbox on CourseLink, named LastName_FirstName_A2.pdf.
- **Generative AI:** Ensure you include the AI coversheet with the assignment
- **Academic integrity:** This is individual work. Submissions are subject to University of Guelph academic misconduct policies.
- **Late Policy:** Refer to the late policy for the course on CourseLink.